

Supplementary V (Cryptanalysis on PF-CLAS)

In Subsection 1.2 of the manuscript, we claim that all existing PF-CLAS schemes have universal algebraic flaws. We classify PF-CLAS schemes up to 2026 in this supplementary, and design two attacks for each category.

1. Scheme Taxonomy

Recently, numerous pairing-free CLAS schemes [1]-[86] have been proposed under existentially unforgeable against chosen-message attacks (EUF-CMA) model. Any such scheme must remain secure against both $\mathcal{A}_1$ and $\mathcal{A}_2$. However, we identify systematic vulnerabilities in the aggregate-verification algorithms of 74 pairing-free CLAS schemes, stemming from incorrect applications of the forking lemma. The other 12 schemes are out of the scope of CLAS. Some schemes also exhibit weaknesses in their individual-verification algorithms, which would necessarily compromise aggregate-verification. For example, Yang et al. [80] have shown that Zhan’s scheme [64] is only vulnerable to an efficient adversary $\mathcal{A}_1$. These issues are isolated and therefore lie beyond the scope of our discussion.

This paper focuses only on the structural defects in the aggregate-verification algorithms and their vulnerabilities to both $\mathcal{A}_1$ and $\mathcal{A}_2$. According to their constructions, existing pairing-free CLAS schemes can be classified into three categories.

- In the verification algorithms of the schemes [1]-[53], only the devices’ public keys are protected by hash values.
- In the verification algorithms of the schemes [54]-[63], only the devices’ ephemeral keys are protected by hash values.
- In the verification algorithms of the schemes [64]-[74], both the devices’ ephemeral keys and public keys are protected by hash values.

All schemes within the same category are vulnerable to the same type of attack. We take the LCPIA scheme [1], Rajkumar’s scheme [57], and Xu’s scheme [67] as examples to present their constructions, the corresponding attacks, and the deficiencies in their formal proofs, respectively. While these three schemes adopt distinct protection strategies in their verification algorithms, they share a fundamental weakness: the aggregate-verification equation does not properly bind aggregated signatures with device identities. Consequently, an adversary can manipulate the aggregation structure and forge valid aggregate signatures. The specific attack methods, however, differ significantly

across the three categories due to their varying levels of protection, as outlined in Table 1:

- **Attack against schemes [1]- [53] (Section 2, exemplified by LCPIA [1]):** The adversary requires only *a single* legitimate device’s full private key to forge aggregate signatures for *any device*, without requiring additional oracle access or secret information of the targeted parties. This attack achieves *universal forgery* with 100% success probability in only 6.12 ms, representing the most severe vulnerability resulting from the fact that only public keys are protected by hash values.
- **Attack against schemes [54]- [63] (Section 3, exemplified by Rajkumar’s scheme [57]):** The adversary requires the full private keys of *multiple* legitimate devices to forge a valid aggregate signature that *includes a single target device* who has never signed the corresponding message. No additional oracle access or secret information of the target device is required; the attack is expected to produce a valid aggregate signature within 36 minutes. The protection of ephemeral keys by hash values mitigates but does not prevent aggregation manipulation.
- **Attack against schemes [64]- [74] (Section 4, exemplified by Xu’s scheme [67]):** The adversary requires the full private keys of *multiple* legitimate devices and access to either `Secret Value` oracle or `Partial Private Key` oracle (to obtain the target device’s secret value or partial private key, respectively). Despite requiring more oracle types, the attack still is expected to produce a valid signature in under 36 minutes. This demonstrates that protecting both public and ephemeral keys with hash values remains insufficient without proper structural binding in the aggregate verification equation.

The following three sections present the constructions, attack procedures, and deficiencies in the formal security proofs of each representative scheme, as well as performance analyses of the corresponding attack variants.

Inductive Validity. The three representative schemes (LCPIA, Rajkumar, Xu) are selected from three exhaustive categories that cover all 74 pairing-free CLAS schemes in the literature. Since (1) every existing scheme falls into exactly one of these three categories based on its verification algorithm structure; (2) all schemes within each category share identical protection mechanisms and thus identical structural vulnerabilities; and (3) we have demonstrated con-

Table 1. COMPARISON OF THE THREE ATTACK METHODS

Schemes	Adversaries	Conditions	Scopes	Time(Security Level)
[1]- [53]	$\mathcal{A}_1$	No target secret is required	The attacker can forge the aggregate signature of any device (multiple devices).	6.12 ms(1-bit)
	$\mathcal{A}_2$			
[54]- [63]	$\mathcal{A}_1$	No target secret is required	The attacker can only forge aggregate signatures involving a single target device.	< 36 min(31-bit)
	$\mathcal{A}_2$			
[64]- [74]	$\mathcal{A}_1$	Only a target secret value is required	The attacker can only forge aggregate signatures involving a single target device.	< 36 min(31-bit)
	$\mathcal{A}_2$	Only a target partial private key is required		

crete attacks on the strongest representative of each category with claimed 128-bit security, the attacks *generalize by structural induction* to all 74 schemes.

Unified Attack Framework. The three attacks instantiate three frameworks that exploits the fundamental structural deficiencies: *the aggregate-verification equation does not bind signatures with ephemeral keys*. The framework adapts to each category by adjusting the collision target—Category 1 targets public key hashes, Category 2 targets ephemeral key hashes, and Category 3 requires oracle assistance to obtain protected components. All three achieve well below 128-bit security (6.12 ms to 36 minutes vs. $> 10^{25}$ years for brute force), demonstrating that the 74 pairing-free CLAS schemes with hash-only protection examined in this work do not achieve their claimed security, across all three protection strategies identified.

2. Analysis of The LCPIA Scheme

This section takes Zhu et al.’s LCPIA scheme [1] as an example to illustrate the vulnerability of these aggregate-verification algorithms [1]- [53], where only devices’ public keys are protected by hash values. The LCPIA scheme presents a pairing-free CLAS scheme.

2.1. Design of LCPIA

This subsection briefly introduces the LCPIA scheme.

- 1) **Setup:** The KGC selects an additive group $\mathbb{G}$ of order p according to the security parameter κ . P is a generator of additive group $\mathbb{G}$. The KGC randomly selects $s \in \mathbb{Z}_p^*$ as its master secret key and calculates $P_{\text{pub}} = sP$. The KGC selects three hash functions: $\mathcal{H}_1, \mathcal{H}_2, \mathcal{H}_3 : \{0, 1\}^* \rightarrow \mathbb{Z}_p^*$. The KGC publishes the public parameter list $(q, P, \mathbb{G}, P_{\text{pub}}, \mathcal{H}_1, \mathcal{H}_2, \mathcal{H}_3)$.
- 2) **Secret Value:** The device U_i randomly chooses $x_i \in \mathbb{Z}_p^*$ as its secret value and calculates $X_i = x_iP$.
- 3) **Partial Private Key:** The KGC randomly chooses $a_i \in \mathbb{Z}_p^*$, computes $A_i = a_iP$, $c_i = a_i + sh_2^i$, where $h_2^i = \mathcal{H}_2(\text{ID}_i, X_i, A_i)$, and sends (A_i, c_i) to U_i securely.

- 4) **Public/Private Key:** U_i sets (x_i, c_i) as its full private key and (X_i, A_i) as its full public key.
- 5) **Signing:** Let m_i be the message. U_i randomly chooses $b_i \in \mathbb{Z}_p^*$, computes ephemeral key $B_i = b_iP$ and integers $h_3^i = \mathcal{H}_3(\text{ID}_i, X_i, B_i, m_i)$ and $z_i = b_i + h_3^i x_i + c_i$, and broadcasts $(\text{ID}_i, X_i, A_i, m_i, B_i, z_i)$.
- 6) **Individual Verification:** Upon receiving $(\text{ID}_i, X_i, A_i, m_i, B_i, z_i)$, the verifier computes $h_2^i = \mathcal{H}_2(\text{ID}_i, X_i, A_i)$ and $h_3^i = \mathcal{H}_3(\text{ID}_i, X_i, B_i, m_i)$, and checks if Equation (1) is valid.

$$z_i P = B_i + h_3^i X_i + A_i + h_2^i P_{\text{pub}} \quad (1)$$

- 7) **Aggregate Verification:** Upon receiving a large number of messages $(\text{ID}_i, X_i, A_i, m_i, B_i, z_i)$, the aggregate-verifier first aggregates the signatures $z_{\text{agg}} = \sum_{i=1}^n z_i$. The aggregate signature is $\sigma = \{z_{\text{agg}}, B_1, B_2, \dots, B_n\}$. It then forwards σ with corresponding messages to the aggregate-verifier. The aggregate-verifier computes $h_2^i = \mathcal{H}_2(\text{ID}_i, X_i, A_i)$ and $h_3^i = \mathcal{H}_3(\text{ID}_i, X_i, B_i, m_i)$, and checks if Equation (2) is valid.

$$z_{\text{agg}} P = \sum_{i=1}^n (B_i + h_3^i X_i + A_i) + \sum_{i=1}^n h_2^i P_{\text{pub}} \quad (2)$$

2.2. Attack $\mathcal{A}_1$ on the LCPIA Scheme

An important contribution of the LCPIA scheme is to design a lightweight aggregate signature without pairing operations, which should be secure against chosen-message attacks (CMA). According to CMA, $\mathcal{A}_1$ does not collude with the KGC. $\mathcal{A}_1$ is assumed to be a legitimate device, which holds at least one full private key. Without loss of generality, let (x_1, c_1) of U_1 be the full private key from oracles Secret Value and Partial Private Key, respectively. $\mathcal{A}_1$ can generate a set of fake identities $\{\text{ID}_i^*\}_{i=2}^n$ to masquerade as a set of devices.

- 1) For $i = 2, \dots, n$, $\mathcal{A}_1$ selects identities ID_i^* , messages m_i^* , random public keys $(X_i^*, A_i^*) \in \mathbb{G}$, and random ephemeral keys $B_i^* \in \mathbb{G}$ and computes $h_2^{i*} = \mathcal{H}_2(\text{ID}_i^*, X_i^*, A_i^*)$, and $h_3^{i*} = \mathcal{H}_3(\text{ID}_i^*, X_i^*, B_i^*, m_i^*)$.

- 2) $\mathcal{A}_1$ selects its own message m_1^* , and computes the aggregate signature $B_1^* = -\sum_{i=2}^n B_i^* - \sum_{i=2}^n (h_3^{i*} X_i^* + A_i^*) - \sum_{i=2}^n h_2^{i*} P_{\text{pub}}$, $h_3^{1*} = \mathcal{H}_3(\text{ID}_1, X_1, B_1^*, m_1^*)$, and $z_{\text{agg}}^* = h_3^{1*} x_1 + c_1$.
- 3) $\sigma^* = (z_{\text{agg}}^*, B_1^*, B_2^*, \dots, B_n^*)$ is the forged aggregate signature. The aggregate-verifier computes h_2^1, h_2^{i*} and h_3^{i*} , and checks Equation (3) to validate σ^* .

$$z_{\text{agg}}^* P = \sum_{i=1}^n B_i^* + \sum_{i=2}^n (h_3^{i*} X_i^* + A_i^*) + \sum_{i=2}^n h_2^{i*} P_{\text{pub}} + h_3^{1*} X_1 + A_1 + h_2^1 P_{\text{pub}} \quad (3)$$

Because $c_1 = a_1 + h_2^1 s$, Equation (3) is valid as shown in Equation (4).

$$\begin{aligned} & \sum_{i=1}^n B_i^* + \sum_{i=2}^n (h_3^{i*} X_i^* + A_i^*) + \sum_{i=2}^n h_2^{i*} P_{\text{pub}} + h_3^{1*} X_1 + A_1 + h_2^1 P_{\text{pub}} \\ &= B_1^* + \sum_{i=2}^n B_i^* + \sum_{i=2}^n (h_3^{i*} X_i^* + A_i^*) + \sum_{i=2}^n h_2^{i*} P_{\text{pub}} + h_3^{1*} X_1 + A_1 + h_2^1 P_{\text{pub}} \\ &= h_3^{1*} X_1 + A_1 + h_2^1 P_{\text{pub}} \\ &= (h_3^{1*} x_1 + a_1 + h_2^1 s) P \\ &= z_{\text{agg}}^* P \end{aligned} \quad (4)$$

$\{\text{ID}_i^*\}_{i=2}^n$ are not identities of legitimate devices. Worse than that, ID_i^* 's corresponding devices may not exist, but the house member has to accept $\{\text{ID}_i^*\}_{i=2}^n$ as the identity of legitimate devices.

2.3. Attack $\mathcal{A}_2$ on the LCPIA Scheme

According to the definition of CMA, $\mathcal{A}_2$ is able to collude with the KGC, but is not able to replace the public keys. $\mathcal{A}_2$ is assumed to be a legitimate device, which holds at least one full private key. Without loss of generality, let (x_1, c_1) of U_1 be the full private key from oracles `Secret Value` and `Partial Private Key`, respectively. $\mathcal{A}_2$ can generate an aggregate signature to frame devices $U_2, U_3, \dots, U_n$.

- 1) According to Subsection 2.1, all devices' identities and public keys (ID_i, A_i, X_i) are public. For $i = 2, \dots, n$, $\mathcal{A}_2$ selects messages m_i^* and random ephemeral keys $B_i^* \in \mathbb{G}$, then computes $h_2^i = \mathcal{H}_2(\text{ID}_i, X_i, A_i)$ and $h_3^{i*} = \mathcal{H}_3(\text{ID}_i, X_i, B_i^*, m_i^*)$.
- 2) $\mathcal{A}_2$ selects its own message m_1^* , and computes the aggregate signature $B_1^* = -\sum_{i=2}^n B_i^* - \sum_{i=2}^n (h_3^{i*} X_i + A_i) - \sum_{i=2}^n h_2^{i*} P_{\text{pub}}$, $h_3^{1*} = \mathcal{H}_3(\text{ID}_1, X_1, B_1^*, m_1^*)$, and $z_{\text{agg}}^* = h_3^{1*} x_1 + c_1$.

$A_i) - \sum_{i=2}^n h_2^{i*} P_{\text{pub}}$, $h_3^{1*} = \mathcal{H}_3(\text{ID}_1, X_1, B_1^*, m_1^*)$, and $z_{\text{agg}}^* = h_3^{1*} x_1 + c_1$.

- 3) $\sigma^* = (z_{\text{agg}}^*, B_1^*, B_2^*, \dots, B_n^*)$ is the forged aggregate signature. The aggregate-verifier computes h_2^i and h_3^{i*} , and checks Equation (5) to validate σ^* .

$$z_{\text{agg}}^* P = \sum_{i=1}^n B_i^* + \sum_{i=1}^n (h_3^{i*} X_i + A_i) + \sum_{i=1}^n h_2^{i*} P_{\text{pub}} \quad (5)$$

Because $c_1 = a_1 + h_2^1 s$, Equation (5) is valid as shown in Equation (6).

$$\begin{aligned} & \sum_{i=1}^n B_i^* + \sum_{i=1}^n (h_3^{i*} X_i + A_i) + \sum_{i=1}^n h_2^{i*} P_{\text{pub}} \\ &= B_1^* + \sum_{i=2}^n B_i^* + \sum_{i=1}^n (h_3^{i*} X_i + A_i) + \sum_{i=1}^n h_2^{i*} P_{\text{pub}} \\ &= -\sum_{i=2}^n B_i^* - \sum_{i=2}^n (h_3^{i*} X_i + A_i) - \sum_{i=2}^n h_2^{i*} P_{\text{pub}} \\ & \quad + \sum_{i=2}^n B_i^* + \sum_{i=1}^n (h_3^{i*} X_i + A_i) + \sum_{i=1}^n h_2^{i*} P_{\text{pub}} \\ &= h_3^{1*} X_1 + A_1 + h_2^1 P_{\text{pub}} \\ &= (h_3^{1*} x_1 + a_1 + h_2^1 s) P \\ &= z_{\text{agg}}^* P \end{aligned} \quad (6)$$

m_i^* is not a message from U_i , but the house member has to accept m_i^* as U_i 's message. $\mathcal{A}_2$ can forge valid aggregate signatures for any number of devices and any message content, without compromising an aggregate-verifier.

2.4. Performance Analysis of the Attacks

We adopt a Type-A elliptic curve [88] and run the implementation on an Ubuntu 24 system equipped with a single-core 1.80-GHz processor and 2-GB of random access memory. NIST explicitly defines 128-bit as the minimum security strength baseline. We conducted 20 repeated runs for each cryptographic operation. The implementation achieves 128-bit security, with execution times of 3.060 ms for a scalar multiplication operation T_m in $\mathbb{G}$, 0.013 ms for a point addition operation in $\mathbb{G}$, 0.001 ms for a general hash operation T_h in $\mathbb{Z}_p^*$. The execution time of modular multiplications and additions is too small to measure. Therefore, the computational cost of point addition, general hash and modular multiplication, and addition operations is negligible. Table 2 presents mean execution time, standard deviation, and 95% confidence interval for T_m and T_h .

From Subsection 2.2 and Subsection 2.3, the computational cost of the attack is $(n+1)T_m$ to generate an aggregate signature to frame n devices. Consequently, we can conclude that $\mathcal{A}_1$ or $\mathcal{A}_2$ can spend $3.060n+3.060$ ms to generate an $n+1$ aggregate signature to frame n devices in the schemes [1]- [53] without exception. The complete source code for all schemes and attacks, along with detailed compilation and running instructions, is provided in Supplementary IV.

Table 2. STATISTICAL ANALYSIS OF CRYPTOGRAPHIC OPERATION TIMES (20 RUNS)

Operation	Mean (ms)	Standard Deviation (ms)	Confidential Interval
T_m	3.060	0.003	[3.054, 3.066]
T_h	0.001	<0.001	[0.001, 0.001]

2.5. Deficiencies in the Formal Security Proof of LCPIA

In the verification algorithms, the ephemeral keys $\{B_i\}_{i=1}^n$ are not protected by any hash values. $\mathcal{A}_1$ or $\mathcal{A}_2$ can freely choose the random ephemeral keys.

Zhu et al. [1] claimed that the LCPIA scheme is unforgeable under the ECDL assumption presented in *Theorem 1* and *Theorem 2* in the LCPIA scheme. However, the LCPIA is proven using the original general forking lemma [89]. If $\mathcal{A}$ can forge a LCPIA signature $(ID_i, X_i, A_i, m_i^*, B_i^*, z_i^*)$, then by rewinding $\mathcal{A}$ with a different random oracle query, a challenger $\mathcal{C}$ can obtain a second forgery $(ID_i, X_i, A_i, m_i^*, B_i^*, z_i^{**})$, and compute $x_i = \frac{z_i^* - z_i^{**}}{h_3^{i*} - h_3^{i**}}$, where both h_3^{i*} and h_3^{i**} are the query results of B_i^* . The equation is valid because $\sigma_i^* = b_i^* + x_i h_3^{i*} + c_i$ and $\sigma_i^{**} = b_i^* + x_i h_3^{i**} + c_i$. Thus, the security of individual LCPIA signatures can be reduced to the hardness of the ECDL assumption.

The original general forking lemma was proposed by Pointcheval et al. for security proofs, especially for pairing-free signature schemes [89], see therein for formal theorems and proofs. The incorrect application of the forking lemma constitutes a fundamental deficiency in the security proofs of all 74 published pairing-free aggregate signature schemes. A complete formal proof of this issue is provided in Supplementary I.

3. Analysis of Rajkumar's Scheme

In this section, we take Rajkumar's scheme as an example to illustrate the vulnerability of these aggregate-verification algorithms [54]- [63], where only devices' ephemeral keys are protected by hash values.

3.1. Design of Rajkumar's Scheme

This subsection briefly introduces Rajkumar's scheme.

- 1) **Set Up, Secret Value, and Public/Private Key:** These algorithms are identical to those in Subsection 2.1.
- 2) **Partial Private Key:** The KGC randomly chooses $a_i \in \mathbb{Z}_p^*$, computes $A_i = a_i P$ and $c_i = a_i + sh_2^i$, where $h_2^i = \mathcal{H}_2(ID_i, A_i)$, then securely returns (A_i, c_i) to U_i .
- 3) **Signing:** Let m_i be the message. U_i randomly chooses $b_i \in \mathbb{Z}_p^*$, computes $B_i = b_i P$, $h_3^i =$

$\mathcal{H}_3(ID_i, X_i, A_i, B_i, m_i)$, and $z_i = h_3^i b_i + x_i + c_i$, and broadcasts $(ID_i, X_i, A_i, m_i, B_i, z_i)$.

- 4) **Individual Verification:** Upon receiving $(ID_i, X_i, A_i, m_i, B_i, z_i)$, the verifier computes $h_2^i = \mathcal{H}_2(ID_i, A_i)$ and $h_3^i = \mathcal{H}_3(ID_i, X_i, A_i, B_i, m_i)$, and checks if Equation (7) is valid.

$$z_i P = h_3^i B_i + X_i + A_i + h_2^i P_{\text{pub}} \quad (7)$$

- 5) **Aggregate Verification:** Upon receiving a large number of messages $(ID_i, X_i, A_i, m_i, B_i, z_i)$, the aggregate-verifier first aggregates the signatures $z_{\text{agg}} = \sum_{i=1}^n z_i$. The aggregate signature is $\sigma = \{z_{\text{agg}}, B_1, B_2, \dots, B_n\}$. It then forwards σ with corresponding messages to the aggregate-verifier. The aggregate-verifier computes $h_2^i = \mathcal{H}_2(ID_i, A_i)$ and $h_3^i = \mathcal{H}_3(ID_i, X_i, A_i, B_i, m_i)$, and checks if Equation (8) is valid.

$$z_{\text{agg}} P = \sum_{i=1}^n (h_3^i B_i + X_i + A_i) + \sum_{i=1}^n h_2^i P_{\text{pub}} \quad (8)$$

3.2. Attack $\mathcal{A}_1$ on Rajkumar's Scheme

$\mathcal{A}_1$ is assumed to be a coalition of legitimate devices. According to the definition of CMA, $\mathcal{A}_1$ can query oracles *Secret Value* and *Partial Private Key* multiple times. Without loss of generality, let (x_i, c_i) of U_i be the full private keys from oracles *Secret Value* and *Partial Private Key*, respectively, where $1 \leq i \leq n$. $\mathcal{A}_1$ can generate a fake identity ID^* to masquerade as a device.

- 1) $\mathcal{A}_1$ selects an identity ID^* , a message m^* , and a public key (A^*, X^*) , and computes $h_2^* = \mathcal{H}_2(ID^*, A^*)$. For $i = 1, 2, \dots, n$, $\mathcal{A}_1$ selects random messages m_i^* and random integers r_i , then computes $B_i^* = r_i P - X^* - A^* - h_2^* P_{\text{pub}}$, and $h_3^{i*} = \mathcal{H}_3(ID_i, X_i, A_i, B_i^*, m_i^*)$, where $\sum_{i=1}^n h_3^{i*} = 1$.
- 2) $\mathcal{A}_1$ selects r^* , and computes $B^* = r^* P$, $h_3^* = \mathcal{H}_3(ID^*, X^*, A^*, B^*, m^*)$, and the aggregate signature $z_{\text{agg}}^* = \sum_{i=1}^n (h_3^{i*} r_i + x_i + c_i) + h_3^* r^*$.
- 3) $\sigma^* = (z_{\text{agg}}^*, B^*, B_1^*, B_2^*, \dots, B_n^*)$ is the forged aggregate signature. The aggregate-verifier computes h_2^* , h_2^i , h_3^* , and h_3^{i*} , and checks Equation (9) to validate σ^* .

$$z_{\text{agg}}^* P = \sum_{i=1}^n (h_3^{i*} B_i^* + X_i + A_i) + \sum_{i=1}^n h_2^i P_{\text{pub}} + h_3^* B^* + X^* + A^* + h_2^* P_{\text{pub}} \quad (9)$$

Because $c_i = a_i + h_2^i s$, Equation (9) is valid as shown

in Equation (10).

$$\begin{aligned}
& \sum_{i=1}^n (h_3^{i*} B_i^* + X_i + A_i) + \sum_{i=1}^n h_2^i P_{\text{pub}} \\
& + h_3^* B^* + X^* + A^* + h_2^* P_{\text{pub}} \\
& = \sum_{i=1}^n h_3^{i*} (r_i P - X^* - A^* - h_2^* P_{\text{pub}}) + \sum_{i=1}^n (X_i + A_i) \\
& + \sum_{i=1}^n h_2^i P_{\text{pub}} + h_3^* B^* + X^* + A^* + h_2^* P_{\text{pub}} \\
& = \sum_{i=1}^n h_3^{i*} r_i P - \sum_{i=1}^n h_3^{i*} (X^* + A^* + h_2^* P_{\text{pub}}) + X^* + \\
& A^* + h_2^* P_{\text{pub}} + \sum_{i=1}^n (x_i + a_i + h_2^i s) P + h_3^* B^* \\
& = \sum_{i=1}^n h_3^{i*} r_i P + \sum_{i=1}^n (x_i + c_i) P + h_3^* r^* P \\
& = z_{\text{agg}}^* P
\end{aligned} \tag{10}$$

ID* is not an identity of a device. Worse than that, ID*'s corresponding device may not exist, but the aggregate-verifier has to accept ID* as the identity of a legitimate device.

3.3. Attack $\mathcal{A}_2$ on Rajkumar's Scheme

$\mathcal{A}_2$ is assumed to be a coalition of legitimate devices. According to the definition of CMA, $\mathcal{A}_2$ can query oracles Secret Value and Partial Private Key multiple times. Without loss of generality, let (x_i, c_i) of U_i be the full private keys from oracles Secret Value and Partial Private Key, respectively, where $2 \leq i \leq n$. $\mathcal{A}_2$ can generate an aggregate signature to frame device U_1 .

- 1) According to Subsection 2.1, U_1 's identity and public key (ID₁, A_1 , X_1) are public. For $i = 1, 2, \dots, n$, $\mathcal{A}_2$ selects random messages m_i^* and random integers r_i , then computes $h_2^1 = \mathcal{H}_2(\text{ID}_1, A_1)$, $B_i^* = r_i P - X_1 - A_1 - h_2^1 P_{\text{pub}}$, and $h_3^{i*} = \mathcal{H}_3(\text{ID}_i, X_i, A_i, B_i^*, m_i^*)$, where $\sum_{i=1}^n h_3^{i*} = 1$.
- 2) $\mathcal{A}_2$ computes the aggregate signature $z_{\text{agg}}^* = \sum_{i=1}^n h_3^{i*} r_i + \sum_{i=2}^n (x_i + c_i)$.
- 3) $\sigma^* = (z_{\text{agg}}^*, B_1^*, B_2^*, \dots, B_n^*)$ is the forged aggregate signature. The aggregate-verifier computes h_2^i and h_3^{i*} , and checks Equation (11) to validate σ^* .

$$z_{\text{agg}}^* P = \sum_{i=1}^n h_3^{i*} B_i^* + \sum_{i=1}^n (X_i + A_i) + \sum_{i=1}^n h_2^i P_{\text{pub}} \tag{11}$$

Because $c_i = a_i + h_2^i s$, Equation (11) is valid as shown in Equation (12).

$$\begin{aligned}
& \sum_{i=1}^n h_3^{i*} B_i^* + \sum_{i=1}^n (X_i + A_i) + \sum_{i=1}^n h_2^i P_{\text{pub}} \\
& = \sum_{i=1}^n h_3^{i*} (r_i P - X_1 - A_1 - h_2^1 P_{\text{pub}}) + X_1 + A_1 \\
& h_2^1 P_{\text{pub}} + \sum_{i=2}^n (X_i + A_i) + \sum_{i=2}^n h_2^i P_{\text{pub}} \\
& = \sum_{i=1}^n h_3^{i*} r_i P - \sum_{i=1}^n h_3^{i*} (X_1 + A_1 + h_2^1 P_{\text{pub}}) + X_1 + \\
& A_1 + h_2^1 P_{\text{pub}} + \sum_{i=2}^n (x_i + a_i + h_2^i s) P \\
& = \sum_{i=1}^n h_3^{i*} r_i P + \sum_{i=2}^n (x_i + c_i) P \\
& = z_{\text{agg}}^* P
\end{aligned} \tag{12}$$

m_1^* is not a message from U_1 , but the aggregate-verifier has to accept m_1^* as U_1 's message. $\mathcal{A}_2$ can forge valid aggregate signatures for any device, without compromising an aggregate-verifier.

3.4. Performance Analysis of the Attacks

The only difference between the LCPIA scheme and Rajkumar's scheme lies in the signing and verification algorithms. In the LCPIA scheme, U_i 's ephemeral key B_i is not protected by any hash function, enabling the efficient chosen-message attack. In Rajkumar's scheme, B_i is protected by h_3^i , yet the scheme remains vulnerable to chosen-message attacks since public-key pair (A_i, X_i) is not protected. The key to the attack is to find a set of hash values $\{h_3^{i*}\}_{i=1}^n$ such that $\sum_{i=1}^n h_3^{i*} = 1$. The problem is equivalent to the generalized birthday problem.

Wagner [90] introduced the Wagner algorithm, an extension of the traditional birthday attack. The Wagner algorithm finds any n inputs $\{x_i\}_{i=1}^n$ such that $\sum_{i=1}^n \mathcal{H}(x_i) = 0$ in $\mathcal{O}(n 2^{\frac{\kappa}{\log \kappa + 1}})$. The Wagner algorithm has been applied to cryptanalysis, e.g., attacking incremental hash functions and signature schemes. The 128-bit security level indicates that an adversary would need to perform approximately 2^{128} fundamental operations (e.g., hash evaluations) to break the scheme via brute force. To achieve 128-bit security, $\kappa = 256$. When $n = 2^{\sqrt{\kappa}-1} = 2^{15}$, the computational cost reaches its minimum $2^{31} T_h$, corresponding to only **31-bit effective security**. The generalized birthday problem and Wagner's algorithm are elaborated in Supplementary II. The simulation results, described in Subsection 2.4, show that the computational cost of a hash evaluation T_h is almost 0.001 ms. It is over 10^{25} years using our laptop to expect to find values x and x' , where $\mathcal{H}(x) = \mathcal{H}(x')$. In contrast, finding a single set of 2^{15} inputs $\{x_i\}_{i=1}^{2^{15}}$, where $\sum_{i=1}^{2^{15}} \mathcal{H}(x_i) = 0$,

requires less than 36 minutes runtime. Consequently, we can conclude that $\mathcal{A}_1$ or $\mathcal{A}_2$ is expected to yield a single forged aggregate signature containing 2^{15} individual signatures within 36 minutes to frame a device in the schemes [54]- [63]. The security of the CLAS schemes [54]- [63] is also based on the original general forking lemma, which is invalid for their schemes. The detailed implementation and raw experimental data of Wagner's algorithm are available in Supplementary III.

4. Analysis of Xu's Scheme

In this section, we take Xu's scheme as an example to illustrate the vulnerability of these aggregate-verification algorithms [64]- [74], where both devices' public keys and ephemeral keys are protected by hash values.

4.1. Design of Xu's Scheme

This subsection briefly introduces Xu's scheme.

- 1) **Setup:** In addition to the setup procedure in Subsection 2.1, the KGC generates and publishes two new hash functions: $\mathcal{H}_4, \mathcal{H}_5 : \{0, 1\}^* \rightarrow \mathbb{Z}_p^*$.
- 2) **Secret Value, and Public/Private Key:** These algorithms are identical to those in Subsection 2.1.
- 3) **Partial Private Key:** The KGC randomly chooses $a_i \in \mathbb{Z}_p^*$, computes $A_i = a_i P$ and $c_i = a_i + sh_2^i$, where $h_2^i = \mathcal{H}_2(\text{ID}_i, X_i, A_i)$, then securely returns (A_i, c_i) to U_i .
- 4) **Signing:** Let m_i be the message. U_i randomly chooses $b_i \in \mathbb{Z}_p^*$, computes $B_i = b_i P$, $h_3^i = \mathcal{H}_3(\text{ID}_i, X_i, A_i, B_i, m_i)$, $h_4^i = \mathcal{H}_4(\text{ID}_i, X_i, A_i, B_i, m_i)$, $h_5^i = \mathcal{H}_5(\text{ID}_i, X_i, A_i, B_i, m_i)$, and $z_i = h_3^i b_i + h_4^i c_i + h_5^i x_i$, and broadcasts $(\text{ID}_i, X_i, A_i, m_i, B_i, z_i)$.
- 5) **Individual Verification:** Upon receiving $(\text{ID}_i, X_i, A_i, m_i, B_i, z_i)$, the verifier computes $h_2^i = \mathcal{H}_2(\text{ID}_i, X_i, A_i)$, $h_3^i = \mathcal{H}_3(\text{ID}_i, X_i, A_i, B_i, m_i)$, $h_4^i = \mathcal{H}_4(\text{ID}_i, X_i, A_i, B_i, m_i)$, and $h_5^i = \mathcal{H}_5(\text{ID}_i, X_i, A_i, B_i, m_i)$, and checks if Equation (13) is valid.

$$z_i P = h_3^i B_i + h_5^i X_i + h_4^i (A_i + h_2^i P_{\text{pub}}) \quad (13)$$

- 6) **Aggregate Verification:** Upon receiving a large number of messages $(\text{ID}_i, X_i, A_i, m_i, B_i, z_i)$, the aggregate-verifier first aggregates the signatures $z_{\text{agg}} = \sum_{i=1}^n z_i$. The aggregate signature is $\sigma = \{z_{\text{agg}}, B_1, B_2, \dots, B_n\}$. It then forwards σ with corresponding messages to the aggregate-verifier. The aggregate-verifier computes $h_2^i = \mathcal{H}_2(\text{ID}_i, X_i, A_i)$, $h_3^i = \mathcal{H}_3(\text{ID}_i, X_i, A_i, B_i, m_i)$, $h_4^i = \mathcal{H}_4(\text{ID}_i, X_i, A_i, B_i, m_i)$, and $h_5^i =$

$\mathcal{H}_5(\text{ID}_i, X_i, A_i, B_i, m_i)$, and checks if Equation (14) is valid.

$$z_{\text{agg}} P = \sum_{i=1}^n (h_3^i B_i + h_5^i X_i + h_4^i A_i + h_4^i h_2^i P_{\text{pub}}) \quad (14)$$

4.2. Attack $\mathcal{A}_1$ on Xu's Scheme

$\mathcal{A}_1$ is assumed to be a coalition of legitimate devices. According to the definition of CMA, $\mathcal{A}_1$ can query oracles Secret Value and Partial Private Key multiple times. Without loss of generality, let (x_i, c_i) of U_i be the full private keys from oracles Secret Value and Partial Private Key, respectively, where $1 \leq i \leq n$. $\mathcal{A}_1$ can generate a fake identity ID^* to masquerade as a device.

- 1) $\mathcal{A}_1$ selects an identity ID^* , a message m^* , two integers $x^*, r^* \in \mathbb{Z}_p^*$ and a public key A^* , and computes $X^* = x^* P$, $h_2^* = \mathcal{H}_2(\text{ID}^*, X^*, A^*)$, $B^* = r^* P$, $h_3^* = \mathcal{H}_3(\text{ID}^*, X^*, A^*, B^*, m^*)$, $h_4^* = \mathcal{H}_4(\text{ID}^*, X^*, A^*, B^*, m^*)$, $h_5^* = \mathcal{H}_5(\text{ID}^*, X^*, A^*, B^*, m^*)$. For $i = 1, 2, \dots, n$, $\mathcal{A}_1$ selects random messages m_i^* and random integers r_i , then computes $B_i^* = r_i P - A^* - h_2^* P_{\text{pub}}$, $h_5^{i*} = \mathcal{H}_5(\text{ID}_i, X_i, A_i, B_i^*, m_i^*)$, $h_4^{i*} = \mathcal{H}_4(\text{ID}_i, X_i, A_i, B_i^*, m_i^*)$, and $h_3^{i*} = \mathcal{H}_3(\text{ID}_i, X_i, A_i, B_i^*, m_i^*)$, where $\sum_{i=1}^n h_3^{i*} = h_3^*$.
- 2) $\mathcal{A}_1$ computes the aggregate signature $z_{\text{agg}}^* = \sum_{i=1}^n (h_3^{i*} r_i + h_5^{i*} x_i + h_4^{i*} c_i) + h_3^* r^* + h_5^* x^*$.
- 3) $\sigma = (z_{\text{agg}}^*, B^*, B_1^*, B_2^*, \dots, B_n^*)$ is the forged aggregate signature. The aggregate-verifier computes h_2^i , h_3^i , h_4^i , h_5^i , h_4^{i*} , h_5^{i*} , and checks Equation (15) to validate σ .

$$z_{\text{agg}}^* P = \sum_{i=1}^n (h_3^{i*} B_i^* + h_5^{i*} X_i + h_4^{i*} A_i + h_4^{i*} h_2^i P_{\text{pub}}) + h_3^* B^* + h_5^* X^* + h_4^* A^* + h_4^* h_2^* P_{\text{pub}} \quad (15)$$

Because $c_i = a_i + h_2^i s$, Equation (15) is valid as shown

in Equation (16).

$$\begin{aligned}
& \sum_{i=1}^n (h_3^{i*} B_i^* + h_5^{i*} X_i + h_4^{i*} A_i + h_4^{i*} h_2^i P_{\text{pub}}) \\
& + h_3^* B^* + h_5^* X^* + h_4^* A^* + h_4^* h_2^* P_{\text{pub}} \\
& = \sum_{i=1}^n h_3^{i*} (r_i P - A^* - h_2^* P_{\text{pub}}) + h_4^* (A^* + h_2^* P_{\text{pub}}) \\
& + \sum_{i=1}^n (h_5^{i*} X_i + h_4^{i*} (A_i + h_2^i P_{\text{pub}})) + h_3^* B^* + h_5^* X^* \\
& = \sum_{i=1}^n (h_3^{i*} r_i P + h_5^{i*} X_i + h_4^{i*} (A_i + h_2^i P_{\text{pub}})) \\
& + h_3^* B^* + h_5^* X^* \\
& = \sum_{i=1}^n (h_3^{i*} r_i + h_5^{i*} x_i + h_4^{i*} (a_i + h_2^i s)) P \\
& + (h_3^* r^* + h_5^* x^*) P \\
& = (\sum_{i=1}^n (h_3^{i*} r_i + h_5^{i*} x_i + h_4^{i*} c_i) + h_3^* r^* + h_5^* x^*) P \\
& = z_{\text{agg}}^* P
\end{aligned} \tag{16}$$

4.3. Attack $\mathcal{A}_2$ on Xu's Scheme

According to the definition of CMA, $\mathcal{A}_2$ has the master secret key s of the KGC, and cannot replace the public keys of the devices. Since $\mathcal{A}_2$ has s , it can query `Partial Private Key` for all devices, including the target device. Without loss of generality, let U_1 be the target device. Additionally, $\mathcal{A}_2$ can query oracle `Secret Value` and obtain secret value x_i , where $2 \leq i \leq n$. Thus, $\mathcal{A}_2$ can generate an aggregate signature to frame device U_1 .

- 1) According to Subsection 2.1, U_1 's identity and public key (ID_1, A_1, X_1) are public. For $i = 1, 2, \dots, n$, $\mathcal{A}_2$ selects random messages m_i^* and random integers r_i , then computes $B_i^* = r_i P - X_1$, $h_5^{i*} = \mathcal{H}_5(\text{ID}_i, X_i, A_i, B_i^*, m_i^*)$, $h_4^{i*} = \mathcal{H}_4(\text{ID}_i, X_i, A_i, B_i^*, m_i^*)$, and $h_3^{i*} = \mathcal{H}_3(\text{ID}_i, X_i, A_i, B_i^*, m_i^*)$, where $\sum_{i=1}^n h_3^{i*} = h_5^{1*}$.
- 2) $\mathcal{A}_2$ computes the aggregate signature $z_{\text{agg}}^* = \sum_{i=1}^n (h_3^{i*} r_i + h_4^{i*} c_i) + \sum_{i=2}^n h_5^{i*} x_i$.
- 3) $\sigma^* = (z_{\text{agg}}^*, B_1^*, B_2^*, \dots, B_n^*)$ is the forged aggregate signature. The aggregate-verifier computes h_2^i , h_3^{i*} , h_4^{i*} and h_5^{i*} , and checks Equation (17) to validate σ^* .

$$z_{\text{agg}}^* P = \sum_{i=1}^n (h_3^{i*} B_i^* + h_5^{i*} X_i + h_4^{i*} A_i + h_4^{i*} h_2^i P_{\text{pub}}) \tag{17}$$

Because $c_i = a_i + h_2^i s$, Equation (17) is valid as shown in Equation (18).

$$\begin{aligned}
& \sum_{i=1}^n h_3^{i*} B_i^* + h_5^{i*} X_i + h_4^{i*} A_i + h_4^{i*} h_2^i P_{\text{pub}} \\
& = \sum_{i=1}^n h_3^{i*} (r_i P - X_1) + h_5^{1*} X_1 + \sum_{i=2}^n h_5^{i*} X_i + \\
& \sum_{i=1}^n h_4^{i*} (A_i + h_2^i P_{\text{pub}}) \\
& = \sum_{i=1}^n h_3^{i*} r_i P - \sum_{i=1}^n h_3^{i*} X_1 + h_5^{1*} X_1 + \sum_{i=2}^n h_5^{i*} X_i + \\
& \sum_{i=1}^n h_4^{i*} (A_i + h_2^i P_{\text{pub}}) \\
& = \sum_{i=1}^n h_3^{i*} r_i P + \sum_{i=2}^n h_5^{i*} x_i P + \sum_{i=1}^n h_4^{i*} (a_i + h_2^i s) P \\
& = \sum_{i=1}^n (h_3^{i*} r_i + h_4^{i*} c_i) P + \sum_{i=2}^n h_5^{i*} x_i P \\
& = z_{\text{agg}}^* P
\end{aligned} \tag{18}$$

4.4. Performance Analysis of the Attacks

Both schemes [1], [57] are vulnerable to forgery attacks: the ephemeral keys B_i are not protected in the LCPIA scheme [1], while the public keys (X_i, A_i) are not protected in Rajkumar's scheme [57]. Both B_i and (X_i, A_i) are protected by hash values in Xu's scheme [67], and Xu et al. claim security via the original general forking lemma. However, the general forking lemma does not directly apply to aggregate signatures, since those hash values are independent.

As in Rajkumar's scheme, Xu's scheme is vulnerable to generalized birthday attacks: $\mathcal{A}_1$ or $\mathcal{A}_2$ can find a set of hash values $\{h_3^{i*} \mid 1 \leq i \leq n\} \cup \{h_5^{1*}\}$ such that $\sum_{i=1}^n h_3^{i*} = h_5^{1*}$. As described in Subsection 3.4, either $\mathcal{A}_1$ or $\mathcal{A}_2$ is expected to generate a valid aggregate signature of 2^{15} signatures to frame a device in less than 36 minutes. This corresponds to 2^{31} hash operations at $T_h = 0.001$ ms per hash, for schemes [64]–[74].

5. Other Schemes

Besides the above three classes, there are also some approaches to construct pairing-free aggregate signatures, but they lie outside our research scope. For example, the schemes [75], [76] are provably secure schemes. However, the definition of those two schemes differs from that of the certificateless signature scheme, and is closer to that of certificate-based signature schemes, as we described in Subsection 1.2 of the manuscript.

The schemes [77]–[82] design pairing-free aggregate signatures. However, from cryptographic perspective,

the aggregate-verifier authenticates the aggregate signature again using a hash message authentication code (HMAC) in their aggregate algorithms. Therefore, those schemes [77]-[82] are not incremental aggregate signatures. In fact, they fall outside the scope of aggregate signatures altogether. Their security relies on HMACs rather than aggregate signature schemes.

The schemes [83] - [86] claimed that they designed CLAS schemes, but their schemes are not aggregate signature schemes; instead, they are typical batch verification schemes, according to the definition in [87].

References

- [1] D. Zhu and Y. Guan, "Secure and Lightweight Conditional Privacy-Preserving Identity Authentication Scheme for VANET," in *IEEE Sensors Journal*, vol. 24, no. 21, pp. 35743-35756, 2024.
- [2] G. K. Verma, S. Rana, A. Tiwari, V. Chamola and N. Kumar, "Dynamic Anonymous Online-Offline Signcryption for Edge-Assisted Internet of Vehicles," in *IEEE Transactions on Vehicular Technology*, vol. 75, no. 1, pp. 1469-1479, 2026.
- [3] I. Ali, Y. Chen, J. Li, A. Wakeel, C. Pan and N. Ullah, "Efficient Offline/Online Heterogeneous-Aggregated Signcryption Protocol for Edge Computing-Based Internet of Vehicles," in *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 12, pp. 14506-14519, 2023.
- [4] W. Wu and J. Chen, "A Security-Enhanced Certificateless Aggregate Signcryption Scheme for Vehicular Ad Hoc Networks", *Vehicular Communications*, vol. 53, 2025.
- [5] C. Yang, L. Tian, X. Wang, F. Lin, R. Jia, Z. Zheng and M. Li, "LAS: Lightweight Aggregate Signcryption for Federated Learning with Blockchain in IoT", *Computer Networks*, vol. 270, 2025.
- [6] B. Wang, X. Guo and S. Zhang, "A Secure and Efficient Pairing-Free Certificateless Aggregate Signcryption Scheme for V2G Networks," in *IEEE Systems Journal*, vol. 18, no. 1, pp. 282-293, 2024.
- [7] W. Wu and F. Ye, "IPCAS: An improved conditional privacy-preserving certificateless aggregate signature scheme without bilinear pairing for VANETs," in *Journal of Systems Architecture*, vol. 152, pp. 1-13, 2024,
- [8] H. Kwon, "Secure and Scalable Device Attestation Protocol with Aggregate Signature," *Symmetry*, vol. 17, no. 5: 698, 2025
- [9] M. A. Faleh, M. Yousif, M. A. Al-Shareeda, M. A. Almaiah, and M. Obaidat, "CSAS-V: Certificateless Schnorr Aggregate Signature for VANETs," in *International Journal of Innovative Research and Scientific Studies*, vol. 8, no. 2, pp. 4000-4011, 2025.
- [10] C. Dai and Z. Xu, "Pairing-Free Certificateless Aggregate Signcryption Scheme for Vehicular Sensor Networks," in *IEEE Internet of Things Journal*, vol. 10, no. 6, pp. 5063-5072, 15 March 15, 2023.
- [11] S. Yu et al., "Efficient ECC-Based Conditional Privacy-Preserving Aggregation Signature Scheme in V2V," in *IEEE Transactions on Vehicular Technology*, vol. 72, no. 11, pp. 15028-15039, Nov. 2023.
- [12] A. Priyadarshini, A. S. Rajasekaran, K. S. Kola, A. Balakrishnan and R. Pattanayak, "An Efficient Signature Aggregation Scheme with Antenna Hardware Implementation for VANETs," in *Scientific Reports* vol. 15, 22784, 2025.
- [13] S. Liu, L. Chen, L. Chen and J. Lai, "Integrated and Accountable Data Sharing for Smart Grids With Fog and Dual-Blockchain Assistance," in *IEEE Transactions on Industrial Informatics*, vol. 20, no. 3, pp. 4940-4952, March 2024.
- [14] C. Wang, Y. Ming, H. Liu, S. Zhang and R. Lu, "Towards Synchronized Privacy-Preserving Authentication for MDTEN-Driven VANETs," in *IEEE Transactions on Vehicular Technology*, vol. 73, no. 3, pp. 4107-4120, 2024.
- [15] K. D. Mukathe, D. Wu and W. Ahmed, "Secure and Efficient Blockchain-Based Certificateless Authentication Scheme for Vehicular Ad-Hoc Networks (VANETs)," in *2022 4th International Conference on Applied Machine Learning (ICAML)*, Changsha, China, pp. 302-307, 2022.
- [16] H. Zhao, X. Zhang, Z. Li, Z. Wei, "An Efficient Pairing-Free Certificateless Signature Scheme with KGC Trust Level 3 for Wireless Sensor Network", in *Wireless Communications and Mobile Computing*, 9884050, 14 pages, 2023.
- [17] L. Meng, N. Deng, Y. Xu and F. Lin, "Reducing In-swarm Verification Cost: A Dynamic Mutual Authentication Scheme for UAV Swarms," in *KSII Transactions on Internet and Information Systems*, vol. 19, no. 5. Korean Society for Internet Information (KSII), 31-May-2025.
- [18] X. Tian, "A Novel Certificateless Aggregate Signature Scheme without Bilinear Pairings", in *Proceedings of the International Conference on Computer Networks and Communication Technology (CNCT 2016)*, pp. 853-857, 2016.
- [19] Y. Deng, S. Wang, Q. Zhang, "Classified data authentication scheme for IoT based on aggregate signature and Hyperledger Fabric," in *Journal of Information Security and Applications*, vol. 85, pp. 1-15, 2024.
- [20] G. Thumbur, G. S. Rao, P. V. Reddy, N. B. Gayathri, D. V. R. K. Reddy and M. Padmavathamma, "Efficient and Secure Certificateless Aggregate Signature-Based Authentication Scheme for Vehicular Ad Hoc Networks," in *IEEE Internet of Things Journal*, vol. 8, no. 3, pp. 1908-1920, 2021.
- [21] Y. Chen and J. Chen, "CPP-CLAS: Efficient and Conditional Privacy-Preserving Certificateless Aggregate Signature Scheme for VANETs," in *IEEE Internet of Things Journal*, vol. 9, no. 12, pp. 10354-10365, 2022.
- [22] D. Lee, K. Yim, I. Lee, "A Certificateless Aggregate Arbitrated Signature Scheme for IoT Environments", in *Sensors*, vol. 20, no. 14, 3983, 2020.
- [23] H. Shu, P. Qi, Y. Huang, F. Chen, D. Xie, and L. Sun, "An Efficient Certificateless Aggregate Signature Scheme for Blockchain-Based Medical Cyber Physical Systems", in *Sensors*, vol. 20, no. 5, 1521, 2020.
- [24] I. A. Kamil and S. O. Ogundoyin, "An improved certificateless aggregate signature scheme without bilinear pairings for vehicular ad hoc networks", in *Journal of Information Security and Applications*, vol. 44, pp. 184-200, 2019.
- [25] L. Zhou, X. Yin, "An improved pairing-free certificateless aggregate signature scheme for healthcare wireless medical sensor networks", *PLOS ONE* vol. 17, no. 7: e0268484, 2022.
- [26] Y. Qu and Q. Mu, "An efficient certificateless aggregate signature without pairing", in *International Journal of Electronic Security and Digital Forensics*, vol. 10, no. 2, pp. 188-203, 2018.
- [27] A. Iqbal, M. Zubair, MA. Khan, I. Ullah, G. Ur-Rehman, AV. Shvetsov, F. Noor, "An Efficient and Secure Certificateless Aggregate Signature Scheme for Vehicular Ad hoc Networks", *Future Internet*, vol. 15, no. 8:266, 2023.
- [28] N. B. Gayathri, G. Thumbur, P. Rajesh Kumar, M. Z. U. Rahman, P. V. Reddy and A. Lay-Ekuakille, "Efficient and Secure Pairing-Free Certificateless Aggregate Signature Scheme for Healthcare Wireless Medical Sensor Networks," in *IEEE Internet of Things Journal*, vol. 6, no. 5, pp. 9064-9075, 2019.
- [29] S. Liu, Y. Liu, W. Liu, Y. Zhang, "A certificateless multi-dimensional data aggregation scheme for smart grid", in *Journal of Systems Architecture*, vol. 140, 2023.
- [30] P. Thorncharoensri, W. Susilo and J. Baek, "Aggregatable Certificateless Designated Verifier Signature," in *IEEE Access*, vol. 8, pp. 95019-95031, 2020,
- [31] S. Jayashree, S.V.N. Santhosh Kumar, "PFCAS—Pairing Free Certificateless Aggregate Scheme for Ensuring Efficient Authentication in Vehicular Ad-hoc Networks," in *Peer-to-Peer Networking and Applications* vol. 17, 3049-3079, 2024.

- [32] Z. Wu, Y. Bai, Y. Zhang, L. Liu, and M. Yue, "TrustCNAV: Certificateless aggregate authentication of civil navigation messages in GNSS," in *Computers & Security*, vol. 148, 2025.
- [33] O.B.J. Rabie, S. Selvarajan, T. Hasanin, et al. "A Full Privacy-preserving Distributed Batch-based Certificateless Aggregate Signature Authentication Scheme for Healthcare Wearable Wireless Medical Sensor Networks (HWMSNs)," in *International Journal of Information Security*, vol. 23, pp. 51–80, 2024.
- [34] S. Jayashree, S.V.N.S. Kumar, "An Efficient Group Signature based Certificateless Verification Scheme for Vehicular Ad-hoc Network," in *Wireless Networks* vol. 30, pp. 3269–3298, 2024.
- [35] Y. Rajkumar, S.V.N.S. Kumar, "A Lightweight Privacy Preserving Distributed Certificateless Aggregate based Mutual Authentication Scheme for Vehicular Ad-hoc Networks," in *Peer-to-Peer Networking and Applications*, vol. 17, pp. 1442–1466, 2024.
- [36] Z. Wang et al., "Key agreement and management scheme based on blockchain for 5G-enabled vehicular networks," in *China Communications*, vol. 22, no. 3, pp. 270–287, 2025.
- [37] Q. Yue, W. Jiang, and H. Lei, "A lightweight certificateless aggregate signature scheme without pairing for VANETs," in *Scientific Reports*, vol. 15, 23663, 2025.
- [38] Z. Ghaleb Al-Mekhlafi et al., "CLA-FC5G: A Certificateless Authentication Scheme Using Fog Computing for 5G-Assisted Vehicular Networks," in *IEEE Access*, vol. 12, pp. 141514–141527, 2024.
- [39] J. Zhang and C. Shi, "An Enhanced-Security Certificateless Aggregate Signcryption for Secure Data Transmission in Resource-Constrained Networks," in *IEEE Internet of Things Journal*, vol. 12, no. 16, pp. 34086–34101, 2025.
- [40] X. Li, X. Yin and J. Ning, "RelCLAS: A Reliable Malicious KGC-Resistant Certificateless Aggregate Signature Protocol for Vehicular Ad Hoc Networks," in *IEEE Internet of Things Journal*, vol. 10, no. 23, pp. 21100–21114, 1 Dec.1, 2023,
- [41] F. Xu and H. Zeng "On the security of two signature schemes for secure communication in IoT environments," in *Multimedia Tools and Applications*, vol. 83, pp. 43673–43683, 2024.
- [42] Y. Zhou et al., "An Efficient Identity Authentication Scheme With Dynamic Anonymity for VANETs," in *IEEE Internet of Things Journal*, vol. 10, no. 11, pp. 10052–10065, 1 June1, 2023.
- [43] Y. Zhou, Z. Wang, Z. Qiao, B. Yang and M. Zhang, "An Efficient and Provably Secure Identity Authentication Scheme for VANET," in *IEEE Internet of Things Journal*, vol. 10, no. 19, pp. 17170–17183, 2023.
- [44] V. Vimal et al., "Certificate-Less Healthcare Signature Scheme for Secure Consumer Technology-Centric Wireless Medical Sensor Devices," in *IEEE Transactions on Consumer Electronics*, vol. 71, no. 2, pp. 5015–5024, May 2025,
- [45] J. Lu, W. Ding, and K. W. Nam, "Smart Meter Data Security Communication Method Based on Bilinear Pairing", in *Proceedings of the 2023 7th International Conference on High Performance Compilation, Computing and Communications*, New York, USA, pp. 201–206, 2023.
- [46] Z. Xu et al., "A Security-Enhanced Conditional Privacy-Preserving Certificateless Aggregate Signature Scheme for Vehicular Ad-Hoc Networks," in *IEEE Internet of Things Journal*, vol. 11, no. 8, pp. 13482–13495, 2024.
- [47] H. Du, Q. Wen and S. Zhang, "An Efficient Certificateless Aggregate Signature Scheme Without Pairings for Healthcare Wireless Sensor Network," in *IEEE Access*, vol. 7, pp. 42683–42693, 2019.
- [48] X. Hu, W. Tan, C. Ma, F. Chen and C. Yu, "Study on Security Analysis and Efficient Improvement of Certificateless Aggregate Signature Scheme," in *2020 IEEE 11th International Conference on Software Engineering and Service Science (ICSESS)*, Beijing, China, 2020, pp. 343–346
- [49] Y. Zhao, Y. Hou, L. Wang, S. Kumari, MK. Khan, H. Xiong, "An efficient certificateless aggregate signature scheme for the Internet of Vehicles", *Transactions Emerging Telecommunications Technologies*, vol. 31:e3708, 2020.
- [50] X. Yang, A. Chen, Z. Wang, X. Du, C. Wang, "Cryptanalysis of an Efficient and Secure Certificateless Aggregate Signature-Based Authentication Scheme for Vehicular Ad Hoc Networks", in *Security and Communication Networks*, 4472945, 12 pages, 2022.
- [51] K. H. Yeh, K. Y. Tsai, and C. Y. Fan, "An efficient certificateless signature scheme without bilinear pairings", in *Multimedia Tools and Applications* vol. 74, pp. 6519–6530, 2015.
- [52] Y. Xie, X. Li, S. Zhang and Y. Li, "iCLAS : An Improved Certificateless Aggregate Signature Scheme for Healthcare Wireless Sensor Networks," in *IEEE Access*, vol. 7, pp. 15170–15182, 2019.
- [53] H. Zheng, M. Luo, Y. Zhang, C. Peng and Q. Feng, "A Security-Enhanced Pairing-Free Certificateless Aggregate Signature for Vehicular Ad-Hoc Networks," in *IEEE Systems Journal*, vol. 17, no. 3, pp. 3822–3833, 2023.
- [54] Y. Zhou, L. Li, X. Zuo and Y. Liu, "A Multiparty Authentication Scheme Based on Aggregate Certificateless Signature for Smart Healthcare," in *IEEE Internet of Things Journal*, vol. 12, no. 17, pp. 35985–36000, 2025.
- [55] L. Cui, Y. Heng, and W. Gang, "Design and Security Analysis of Certificateless Aggregate Signature Scheme", in *Artificial Intelligence and Security. ICAIS 2019*, vol 11635, 2019.
- [56] A. Tomar, S. Tripathi and K. Arivarasan, "A Blockchain-Based Certificateless Aggregate Signature Scheme for Fog-Enabled Smart Grid Environment," in *IEEE Transactions on Green Communications and Networking*, vol. 7, no. 4, pp. 1892–1905, Dec. 2023.
- [57] Y. Rajkumar, S. V. N. S. Kumar, "An Elliptic Curve Cryptography based Certificate-less Signature Aggregation Scheme for Efficient Authentication in Vehicular Ad Hoc Networks," in *Wireless Networks* vol. 30, pp. 335–362, 2024.
- [58] X. Ye, G. Xu, X. Cheng, Y. Li, Z. Qin, "Certificateless-Based Anonymous Authentication and Aggregate Signature Scheme for Vehicular Ad Hoc Networks", in *Wireless Communications and Mobile Computing*, 6677137, 16 pages, 2021
- [59] K. Li, M.H. Au, W. H. Ho, Y. L. Wang, "An Efficient Conditional Privacy-Preserving Authentication Scheme for Vehicular Ad Hoc Networks Using Online/Offline Certificateless Aggregate Signature", In: *Provable Security. ProvSec 2019*, Lecture Notes in Computer Science(), vol 11821. Springer, Cham.
- [60] J. Cui, J. Zhang, H. Zhong, R. Shi, Y. Xu, "An efficient certificateless aggregate signature without pairings for vehicular ad hoc networks", in *Information Sciences*, vol. 451–452, pp. 1–15, 2018.
- [61] S. Zhang, M. Ma, and B. Wang, "A Lightweight Privacy Preserving Scheme of Charging and Discharging for Electric Vehicles based on Consortium Blockchain in Charging Service Company", in *International Journal of Electrical Power & Energy Systems*, vol. 143, 108499, 2022.
- [62] G. Verma, A. Tiwari, M. Wadhwa and N. Kumar, "PUF Enabled and Dynamic Anonymous Certificateless Batch-Verifiable Signcryption for IoMT", *Computers and Electrical Engineering*, vol. 128, Part A, 2025.
- [63] A. Bansal, S. Kumari, N. Doshi, M. Amoon and M. Hölbl, "Security Enhanced CLS and CL-AS Scheme Without Pairings for VANETs," in *IEEE Access*, vol. 13, pp. 42508–42520, 2025.
- [64] Y. Zhan, B. Wang and R. Lu, "Cryptanalysis and Improvement of a Pairing-Free Certificateless Aggregate Signature in Healthcare Wireless Medical Sensor Networks," in *IEEE Internet of Things Journal*, vol. 8, no. 7, pp. 5973–5984, 2021.
- [65] G. Verma, N. Wani, S. Rana, N. Kumar and A. Tiwari, "Dynamic-Anonymous Privacy-Preserving Authenticated Aggregation for Safety-Warning System for the Internet of Vehicles", *Journal of Information Security and Applications*, vol. 84, 2024.

- [66] B. Liu, X. Yao, K. Guo and P. Zhu, "Consortium Blockchain Based Lightweight Message Authentication and Auditing in Smart Home," in *IEEE Access*, vol. 11, pp. 68473-68485, 2023.
- [67] R. Xu, Y. Zhou, Q. Yang, K. Yang, Y. Han, B. Yang, and Z. Xia, "An efficient and secure certificateless aggregate signature scheme", *Journal of Systems Architecture*, vol. 147: 103030, 2024.
- [68] X. Liu, Y. Zhou, Y. Zhu, Z. Xia, "A novel construction of certificateless aggregate signcryption scheme for smart healthcare", *Expert Systems with Applications*, vol. 285: 127963, 2025.
- [69] H. Yu, J. Zhou and M. Ma, "Anonymous Batch Message Authentication Aided by Edge Servers in Industrial Internet of Thing," in *2024 IEEE World AI IoT Congress (AIoT)*, Seattle, WA, USA, pp. 584-590, 2024.
- [70] K. -A. Shim, "Cryptanalysis of Compact Certificateless Aggregate Signature Schemes for HWMSNs and VANETs," in *IEEE Access*, vol. 12, pp. 137634-137641, 2024.
- [71] Z. Shen, X. Kou, T. Yang, H. Xu, D. Wang and S. Niu, "A Revocable Pairing-Free Certificateless Signature Scheme Based on RSA Accumulator," in *23rd International Conference on Trust, Security and Privacy in Computing and Communications*, Sanya, China, pp. 149-156, 2024.
- [72] Z. Gong, T. Gao, N. Guo, "PCAS: Cryptanalysis and improvement of pairing-free certificateless aggregate signature scheme with conditional privacy-preserving for VANETs", in *Ad Hoc Networks*, vol. 144, 2023.
- [73] W. Yang, T. Li, F. Zhang, Y. Zheng and Z. Liu, "Practical Certificateless Aggregate Signcryption With Public Verification for IoMT," in *IEEE Internet of Things Journal*, vol. 13, no. 3, pp. 5266-5278, 2026.
- [74] Z. Ghaleb Al-Mekhlafi et al., "CLA-FC5G: A Certificateless Authentication Scheme Using Fog Computing for 5G-Assisted Vehicular Networks," in *IEEE Access*, vol. 12, pp. 141514-141527, 2024.
- [75] S. Pan, Y. Zhang and S. Wang, "A Secure Aggregate Authentication Scheme With Efficient Revocation for IoT-Based Primary
- [76] Z. Sui, Y. Zha, Y. Sun and M. Duan, "FB-ASC+: Improved Fog-Blockchain Assisted Certificateless Aggregate Signcryption Scheme," in *IEEE Transactions on Industrial Informatics*, (early access).
- [77] W. Yang, S. Wang and Y. Mu, "An Enhanced Certificateless Aggregate Signature Without Pairings for E-Healthcare System," in *IEEE Internet of Things Journal*, vol. 8, no. 6, pp. 5000-5008, 2021.
- [78] W. Xiong, R. Wang, Y. Wang, Y. Wei, F. Zhou and X. Luo, "Improved Certificateless Aggregate Signature Scheme Against Collusion Attacks for VANETs," in *IEEE Systems Journal*, vol. 17, no. 1, pp. 1098-1109, 2023.
- [79] L. Deng, Y. Yang and R. Gao, "Certificateless Designated Verifier Anonymous Aggregate Signature Scheme for Healthcare Wireless Sensor Networks," in *IEEE Internet of Things Journal*, vol. 8, no. 11, pp. 8897-8909, 2021.
- [80] X. Yang, H. Wen, R. Diao, X. Du and C. Wang, "Improved Security of a Pairing-Free Certificateless Aggregate Signature in Healthcare Wireless Medical Sensor Networks," in *IEEE Internet of Things Journal*, vol. 10, no. 12, pp. 10881-10892, 2023.
- [81] X. Yang, S. Li, L. Yang, X. Du and C. Wang, "Efficient and Security-Enhanced Certificateless Aggregate Signature-Based Authentication Scheme With Conditional Privacy Preservation for VANETs," in *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 9, pp. 12256-12268, 2024.
- [82] W. Wu and F. Ye, "A Secure and Efficient Certificateless Aggregate Signature Authentication Scheme With Pseudonyms for VANETs," in *IEEE Internet of Things Journal*, vol. 12, no. 1, pp. 124-139, 2025.
- [83] R. Guo, R. Dong, X. Li, Y. Zhang and D. Zheng, "DRCLAS: An efficient certificateless aggregate signature scheme with dynamic revocation in vehicular ad-hoc networks," in *Vehicular Communications*, vol. 47, pp. 1-15, 2024.
- [84] D. Chen, Fei Zhou, Yining Liu, Lihui Li, Yangfan Liang, "Secure pairing-free certificateless aggregate signcryption scheme for IoT," in *Journal of Systems Architecture*, vol. 156, 103268, 2024.
- [85] Y. Wang, C. Peng, X. Jia, J. Wen and Y. Zhang, "Pairing-Free Blockchain-Assisted Certificateless Aggregation Signcryption Scheme for VANETs," in *IEEE Internet of Things Journal*, vol. 12, no. 11, pp. 15545-15557, 2025.
- [86] Z. Wang, H. Wang, Y. Wang, and X. Yang, "CLASRM: A Lightweight and Secure Certificateless Aggregate Signature Scheme with Revocation Mechanism for 5G-Enabled Vehicular Networks", in *Wireless Communications and Mobile Computing*, 3646960, 20 pages, 2022.
- [87] J. Camenisch, A. Lysyanskaya, "Signature schemes and anonymous credentials from bilinear maps", in *Advances in Cryptology — CRYPTO 2004*, vol. 3152, pp. 56–72. Springer, 2004.
- [88] B. Lynn, "Pbc Library", 3rd ed. Online: <http://crypto.stanford.edu/pbc>, 2006.
- [89] D. Pointcheval and J. Stern, "Security arguments for digital signatures and blind signatures," in *Journal of Cryptology*, vol. 13, no. 3, pp. 361–396, 2000.
- [90] D. Wagner, "A Generalized Birthday Problem," in *Advances in Cryptology — CRYPTO 2002. CRYPTO 2002, Lecture Notes in Computer Science*, vol. 2442, pp 288–304, 2002.